

Project Proposal

Project #2 - DBS Electrode Precise Localization in the Brain

Submission Information

Student 1	Samad Ullah Email: samad.ullah@rai.usc.gal
Student 2	Catarina Souto Email: up202502447@up.pt
Institutions	University of Santiago de Compostela University of Porto
Submitted to	Professor Joao Paulo Cunha

Project Overview

Deep Brain Stimulation (DBS) is a neurosurgical treatment in which electrodes are implanted in specific brain regions to treat neurological disorders such as Parkinson's disease, essential tremor, and dystonia. This project proposes a workflow for precise DBS electrode localization by combining pre-operative MRI and post-operative CT. MRI provides detailed anatomical information, whereas CT offers clear visualization of the implanted electrodes. By registering both modalities into a common spatial framework, the electrode position can be interpreted accurately with respect to the underlying brain anatomy.

1. Main Objective

The main objective of this project is to develop a biomedical image analysis workflow capable of registering post-operative CT to pre-operative MRI in order to localize DBS electrodes accurately within anatomical brain space. This fused representation will support anatomically meaningful post-operative verification of implantation accuracy.

2. Proposed Methodology

The proposed workflow is organized into five technical stages:

- **Data inspection and preprocessing:** Inspection of image dimensions, voxel spacing, orientation, and overall data quality, followed by preprocessing steps such as resampling,

normalization, cropping around the brain region, and optional denoising or bias correction where appropriate.

- **Multimodal image registration:** Registration of post-operative CT to pre-operative MRI using a multimodal approach, initially based on rigid registration. Mutual information will be considered as the main similarity metric due to its suitability for MRI-CT alignment.
- **Image fusion and visualization:** Generation of fused MRI-CT visualizations to assess alignment quality and facilitate anatomical interpretation in axial, sagittal, and coronal views.
- **Electrode localization:** Identification of the DBS electrode in the post-operative CT and mapping of its position into MRI space for precise anatomical localization relative to visible brain structures.
- **Evaluation:** Assessment of the workflow through qualitative inspection of fused images and, where feasible, quantitative measures related to registration quality and localization precision.

3. Scientific and Clinical Relevance

This project addresses a clinically meaningful problem in neuroimaging and image-guided neurosurgery. Accurate electrode localization is essential for evaluating implantation success and understanding the relationship between electrode position and therapeutic outcome. From a biomedical image analysis perspective, the project involves multimodal image registration, medical image fusion, and spatial localization of implanted devices in a clinically relevant context.

4. Expected Outcomes

The expected outcome is a coherent workflow for aligning pre-operative MRI and post-operative CT, generating fused visualizations, and localizing DBS electrodes accurately with respect to patient-specific brain anatomy. The final result should demonstrate both technical validity and clinical interpretability.

5. Concluding Statement

In summary, this proposal aims to address the challenge of DBS electrode localization through the fusion of complementary neuroimaging modalities. By combining the anatomical detail of MRI with the implant visibility of CT, the project seeks to produce a practical, rigorous, and clinically relevant solution grounded in biomedical image analysis methods.

Submitted by

Samad Ullah University of Santiago de Compostela samad.ullah@rai.usc.gal	Catarina Souto University of Porto up202502447@up.pt
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